



Petroleum Exploration Economics & Decision Strategies



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While many wildcat wells result in dry holes, those that succeed can transform corporate trajectories and redefine industry dynamics. Hydrocarbon exploration is inherently high-risk and capital-intensive, with significant uncertainties ranging from geological variability to fluctuating market conditions.

This course provides participants with advanced decision-analysis frameworks and economic valuation techniques, so that they effectively evaluate exploration opportunities and make informed decisions. This program emphasizes value-driven decision-making, integrating probability modeling, subsurface volume uncertainty, risk-adjusted financial assessment, and uncertainty management.



LOCATION

Rio de Janeiro

London

Singapore

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COURSE OBJECTIVES

We will cover the art and science of valuation in hydrocarbon exploration. At the end of this course, the participants will learn to:

- Define the concept of economic value in hydrocarbon exploration and its role in investment decisions.
- Explain the sources of uncertainty in exploration, including geological, technical, operational, market, and regulatory risks.
- Apply decision-analysis frameworks, including decision trees, scenario analysis, and sensitivity analysis, to evaluate exploration opportunities.
- Analyze probabilistic resource assessments and reserve-based valuation models to optimize exploration strategies.
- Evaluate investment risks using Monte Carlo simulation techniques and uncertainty quantification methodologies.
- Assess exploration prospects and conduct economic feasibility studies to inform investment decisions.
- Develop risk mitigation strategies based on industry's best practices and emerging valuation techniques.
- Formulate strategic recommendations for value creation from exploration projects.



This course is ideal for:

- Petroleum geologists, geoscientists, and explorationists.
- Energy analysts, project managers, and financial specialists.
- Petroleum engineers, facilities engineers, and technical staff who are interested in exploration valuation and decision making.
- Business development strategists, investment analysts, and risk management experts.
- Market analysts and forecasters.
- supply chain professionals, procurement specialists, regulatory advisors.

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Day 1

Drilling for Success

- The petroleum exploration value chain: from prospect generation to field development.
- Strategic commercial considerations in exploration projects.
- Global energy trends: future market conditions shaping petroleum investments.
- Economic decision-making for value creation: beyond volume-based assessments.
- Decision analysis, economic principles, and uncertainty: understanding expected value.
- Case Study: Drill a wildcat well or sell the exploration license?

Day 2

Techno-Economic Valuations

- Probability-based exploration models: encoding geological uncertainty.
- Monte Carlo simulation for hydrocarbon reserves analysis.
- The role of price and technology in conventional vs. non-conventional reserves.
- Financial valuation in upstream petroleum projects.
- Portfolio effects in multi-prospect plays: risk diversification in exploration strategy.

Day 3

Market Forces & Financial Risk Considerations

- Hydrocarbon price volatility and corporate price forecasts.
- Market risk and discounting principles for exploration investments.
- The impact of time on risk-adjusted valuations.
- Introduction to the Capital Asset Pricing Model (CAPM) and its relevance to exploration valuation
- Case Studies: Appraisal well valuation, sequential exploration strategy, and waiting options.



Day 4

Analytical tools for Exploration Decision Support

- Decision Trees for exploration modeling: structuring sequential investment decisions.
- Conditional probabilities in drilling outcomes and reserves estimation.
- Bayesian theorem applications in updating exploration success probabilities.
- Market price forecasting and its role in economic valuations.
- Value of Information (VOI) analysis for exploration efficiency.
- Case Study: Assessing the economic value of seismic data before drilling.

Day 5

Future Industry Trends & Strategic Positioning

- Long-term success in exploration: portfolio diversification strategies.
- Environmental considerations and emission trading schemes in petroleum investments.
- Case Study: Exploration investment decisions under carbon pricing regulations.
- Case Study: Optimizing sequential exploration in multi-prospect plays.
- Concluding remarks: Does “fortune favor the bold,” or is systematic risk analysis the key to success?





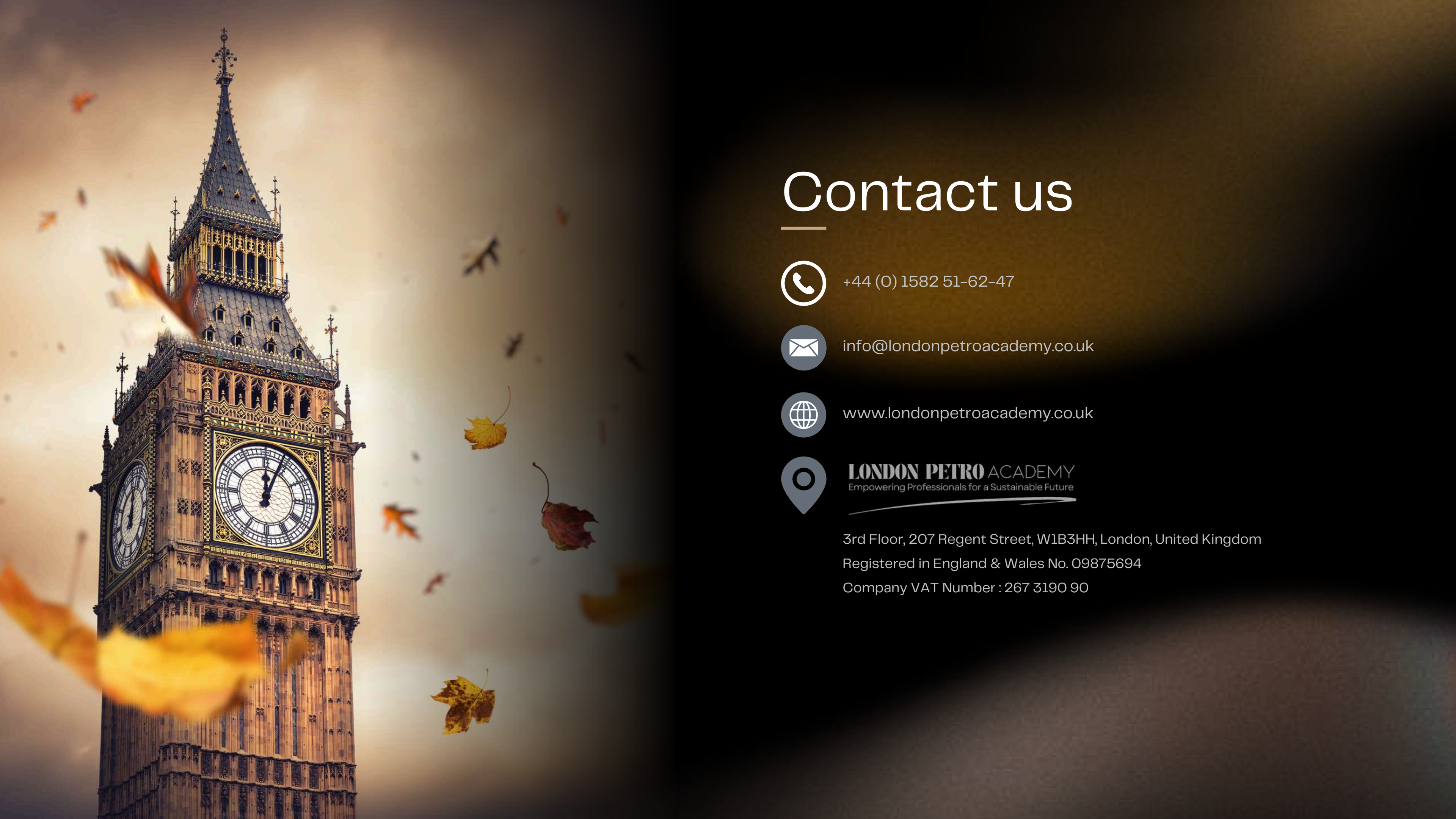
BABAK JAFARIZADEH

Dr. Babak Jafarizadeh is a senior consultant and engineering economist who views engineering as the pursuit of economic solutions to technical challenges in an uncertain world. With extensive experience in the oil and gas industry, he has served as a senior economic analyst supporting multi-million-dollar investment decisions for projects across the globe.

Since 2019, Dr. Jafarizadeh has worked as a senior consultant for leading training and consulting companies, while also holding academic appointments at several prestigious institutions — including Heriot-Watt University in the UK (Assistant Professor), the University of California, Berkeley (Lecturer at the Fung Institute for Engineering Leadership), and previously the University of Greenwich (Senior Lecturer). His earlier industry experience includes serving as a Senior Economic Analyst with Equinor (formerly Statoil) in Norway and the USA.

Dr. Jafarizadeh holds a PhD in Petroleum Engineering with a concentration in Investment and Decision Analysis from the University of Stavanger (in collaboration with the University of Texas at Austin), as well as an MSc in Financial Engineering and a BSc in Industrial Engineering. He is an active member of several professional associations, including the Society of Petroleum Engineers (SPE), the Institute for Operations Research and the Management Sciences (INFORMS), and the European Association of Geoscientists and Engineers (EAGE).

His teaching and research interests lie at the intersection of decision analysis and corporate finance, with particular applications in engineering and the energy industry. He has designed and delivered numerous professional training courses, such as Advanced Financial Modelling for Oil and Gas Professionals, Advanced Project Economics, Oil and Gas: Markets, Trading, Pricing and Economic Frameworks, and Energy Economics. Dr. Jafarizadeh is also the author of the book Economic Decision Analysis (for Project Feasibility Studies) and continues to contribute to both academia and industry through research, consulting, and executive education. His areas of expertise include petroleum economics, investment and decision analysis, project feasibility studies, corporate finance, and financial modelling for energy projects.



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